AI-Based Spatio-temporal Urban Growth Analysis using NDBI and Machine Learning

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Winter Training Program on Space Science Technology

Remote Sensing and GIS

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1. TITLE

AI-Based Spatio-temporal Urban Growth Analysis using NDBI and Machine Learning

2. OBJECTIVE

The primary objective of this research project is to comprehensively analyse, quantify, and visualize the spatiotemporal dynamics of urban expansion within the Greater Hyderabad Metropolitan Region over a 24-year observational period (2000–2024). To achieve this, the study employs advanced remote sensing techniques and multispectral satellite imagery, specifically leveraging the Normalized Difference Built-up Index (NDBI) in conjunction with the Normalized Difference Vegetation Index (NDVI). This dual-index approach is utilized to accurately isolate concrete infrastructure from complex environmental noise, such as highly reflective barren agricultural soil and atmospheric inconsistencies.

Furthermore, this project aims to bridge the critical gap between traditional geographic spatial mapping and predictive data science. By addressing historical satellite anomalies, such as sensor hardware failures and cloud cover limitations, through a rigorous methodology—integrating custom QGIS raster thresholding with algorithmic NDVI masking—the study ensures strict quantitative accuracy in its square-kilometre area extractions across all observation years.

Finally, the extracted geographic information system (GIS) statistics are integrated into a Python-based Machine Learning environment. By training a supervised Simple Linear Regression model on the historical baseline data, the ultimate goal is to model past developmental trajectories and mathematically forecast the total built-up area extent for the year 2030. This predictive intelligence is designed to provide actionable, data-driven insights for urban planners, real estate developers, and policymakers tasked with managing rapid infrastructure sprawl and sustainable metropolitan development.

3. STUDY AREA

The selected Area of Interest (AOI) for this analysis is the Greater Hyderabad Metropolitan Region, located in the state of Telangana, India. To accurately capture the true scale of suburban sprawl and IT-sector expansion, the geographic boundaries of two adjacent districts – Hyderabad District and Ranga Reddy District were merged to form a comprehensive study area.

Hyderabad was selected due to its rapid transformation into a tier-1 global IT hub over the last two decades. The historic core (Hyderabad District) was largely saturated with built-up infrastructure prior to the year 2000. Therefore, incorporating the Ranga Reddy district into the AOI was essential to capture the massive horizontal urban expansion corridors, specifically towards the western zones such as HITEC city, Gachibowli and the Financial District.

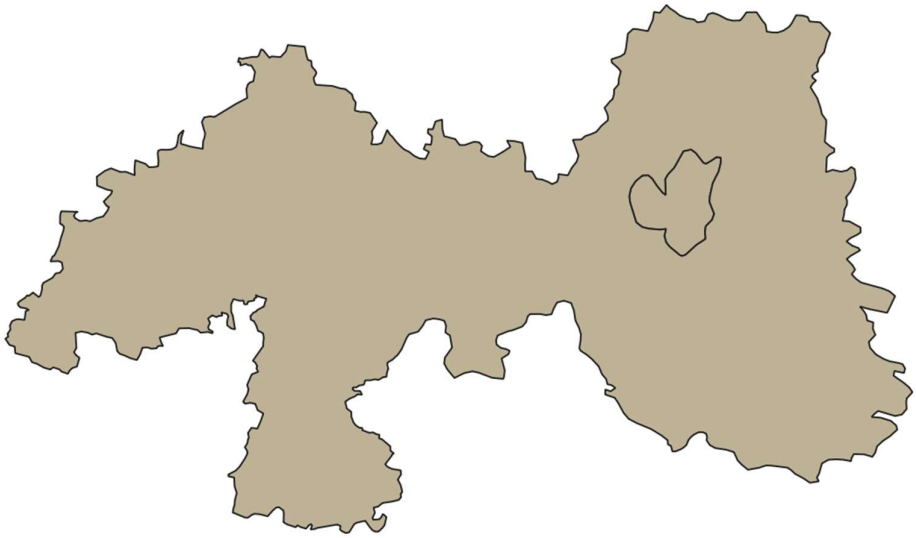

Figure 1: Geographic boundary of the Greater Hyderabad Study Area (Merging Hyderabad and Ranga Reddy districts), extracted from GADM shapefile data.

4. DATA USED

To ensure an accurate time-series analysis over a 24-year period, multispectral satellite imagery was acquired from the United States Geological Survey (USGS) Earth Explorer portal. The dataset consists of Landsat scenes captured at approximately 5-year intervals (2000, 2010, 2015, 2020, and 2024)[1].

To maintain consistency and minimize atmospheric interference, all selected scenes were acquired with minimal cloud cover over the Greater Hyderabad study area in the months of September/October.

Year	Satellite/Sensor	Collection Level	Bands Used for NDBI
2000	Landsat 7 (ETM+)	Collection 2 Level 2	Band 4 (NIR) & Band 5 (SWIR 1)
2010	Landsat 4-5 (TM)	Collection 2 Level 2	Band 4 (NIR) & Band 5 (SWIR 1)
2015	Landsat 8-9 (OLI/TIRS)	Collection 2 Level 2	Band 5 (NIR) & Band 6 (SWIR 1)
2020	Landsat 8-9 (OLI/TIRS)	Collection 2 Level 2	Band 5 (NIR) & Band 6 (SWIR 1)
2024	Landsat 8-9 (OLI/TIRS)	Collection 2 Level 2	Band 5 (NIR) & Band 6 (SWIR 1)

Table 1: Details of Landsat Imagery acquired from USGS Earth Explorer for the Greater Hyderabad Study Area

Software and Tools used:

- QGIS (Quantum GIS): Utilised for raster clipping, band calculation (Raster Calculator), vectorization, and generating final map layouts.
- Python: Utilised for data processing (Pandas, NumPy) and Machine Learning (Scikit-Learn for Linear Regression).
- Microsoft Excel: Utilised for tabular data organization and preliminary statistical analysis.

[1] *Note: The year 2005 was excluded from the 5-year interval sequence due to the Landsat 7 Scan Line Corrector (SLC) failure that occurred in 2003, which resulted in significant data gaps (striping) over the study area. Furthermore, suitable cloud-free Landsat 5 TM data was unavailable for this specific region during the 2004–2006 window.*

5. METHODOLOGY

The methodology for this time-series analysis was executed in four primary phases: spatial data preprocessing, spectral index computation (NDBI and NDVI), mathematical thresholding/masking to isolate urban features, and quantitative area extraction followed by Machine Learning forecasting.

5.1 Data Preprocessing and AOI Extraction

The raw Landsat imagery acquired from Earth Explorer consisted of large-scale raster data. Processing the entire scene is computationally expensive and unnecessary for localised urban analysis. Therefore, the first step involved extracting the Region of Interest (ROI).

Using QGIS, the global administrative boundary shapefiles (GADM) were filtered to isolate the Hyderabad and Ranga Reddy district polygons. These vector layers were merged to create a unified AOI mask. Using the "Clip Raster by Mask Layer" processing algorithm in QGIS, the raw Near-Infrared (NIR) and Shortwave-Infrared (SWIR) raster bands for all five observation years (2000, 2010, 2015, 2020, 2024) were precisely clipped to the Greater Hyderabad boundary.

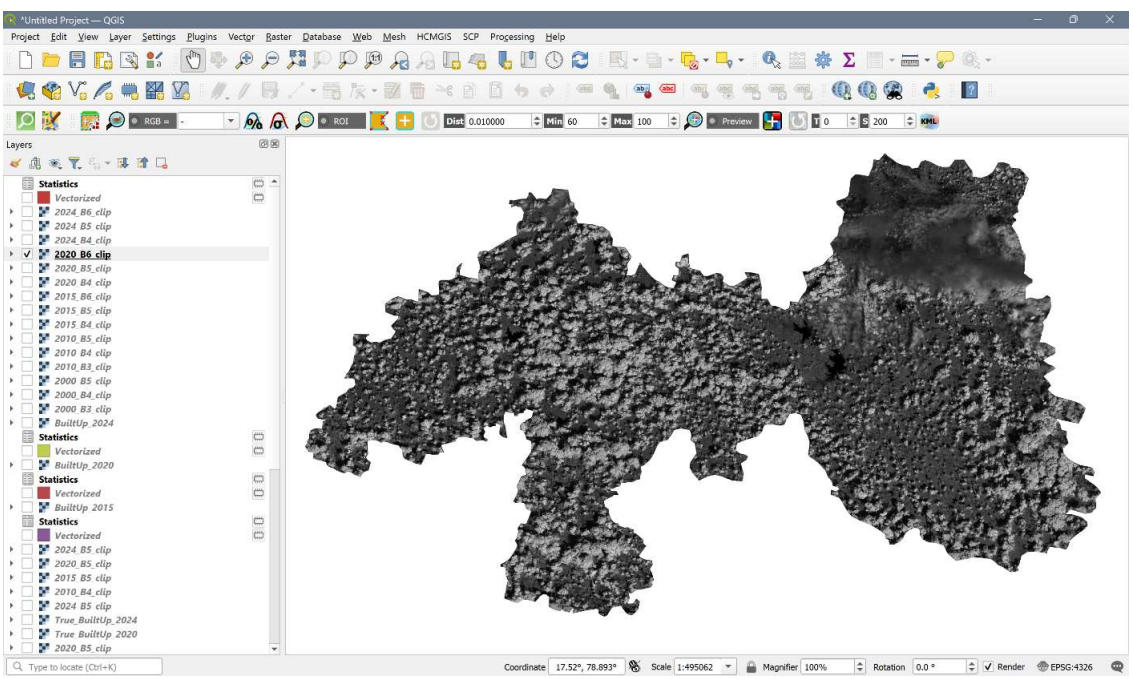

Figure 2: Execution of the Raster Clipping in QGIS to isolate the Greater Hyderabad Study Area for the year 2020 and band B6.

5.2 Normalized Difference Built-Up Index (NDBI) Computation

To automatically detect and map urban infrastructure (buildings, roads and concrete surfaces), the NDBI spectral index was utilised. NDBI exploits the unique spectral response of urban areas, which reflect Shortwave-Infrared (SWIR) light highly but absorb Near-Infrared (NIR) light.

The NDBI was computed using the Raster Calculator in QGIS. Because the dataset spans different Landsat sensor generations, the specific band numbers utilised in the formula varied to ensure spectral consistency:

For Landsat 7 (2000) and Landsat 4-5 (2010):

$$((\boldsymbol{Band\ 5} - \boldsymbol{Band\ 4})\ /\ (\boldsymbol{Band\ 5} + \boldsymbol{Band\ 4})) > \mathbf{0}$$

For Landsat 8-9 (2015, 2020, 2024):

$$((\boldsymbol{Band\ 6} - \boldsymbol{Band\ 5})\ /\ (\boldsymbol{Band\ 6} + \boldsymbol{Band\ 5})) > \mathbf{0}$$

By appending > 0 to the equation, the QGIS Raster Calculator instantly generated a binary output raster where pixel values greater than zero (built-up land) were assigned a value of 1, and all other land covers were assigned 0.

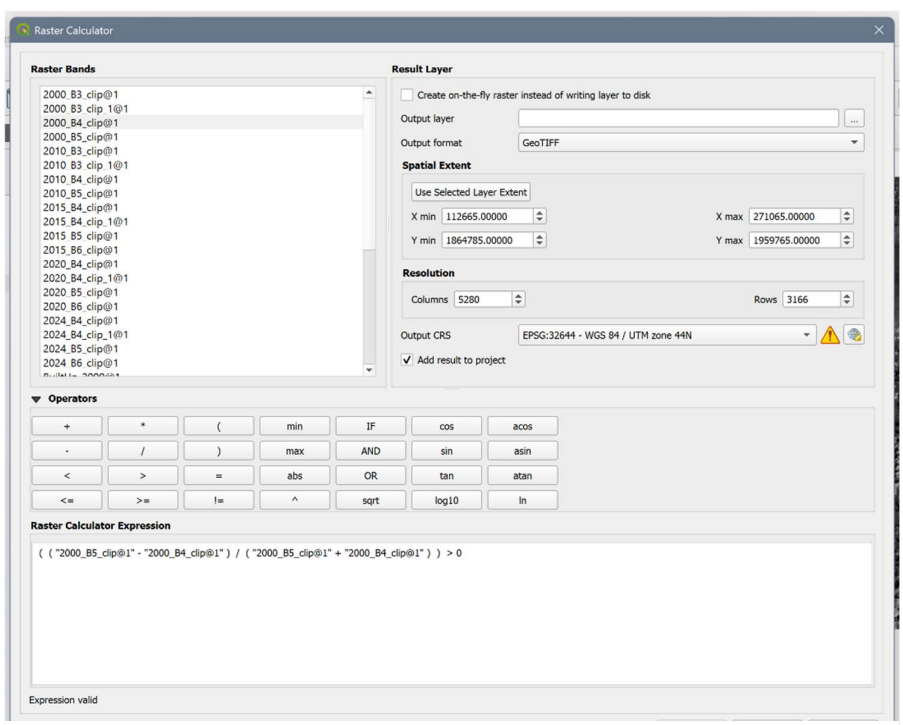

Figure 3: NDBI computation and binary thresholding for the year 2000 utilising the QGIS Raster Calculator expression builder.

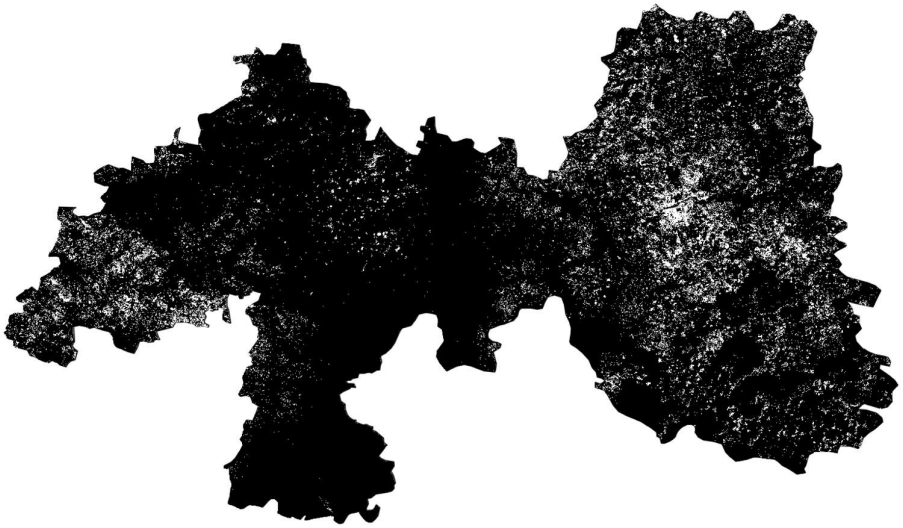

Figure 4: Binary NDBI built-up extraction map for the Greater Hyderabad study area, year 2020.

5.3 Algorithmic Bare-Soil Masking using NDVI.

A known limitation of the NDBI algorithm is its tendency to confuse highly reflective bare soil or dry agricultural land with built-up concrete surfaces. To eliminate these "false positive" urban pixels, the Normalized Difference Vegetation Index (NDVI) was introduced as a secondary algorithmic mask.

Using the Raster Calculator, the NDVI was calculated utilizing the Red and Near-Infrared (NIR) bands. To calculate the NDVI layer:

For Landsat 7 (2000) and Landsat 4-5 (2010):

$$((\boldsymbol{Band\ 4} - \boldsymbol{Band\ 3})\ /\ (\boldsymbol{Band\ 4} + \boldsymbol{Band\ 3}))$$

For Landsat 8-9 (2015, 2020, 2024):

$$((\boldsymbol{Band\ 5} - \boldsymbol{Band\ 4})\ /\ (\boldsymbol{Band\ 5} + \boldsymbol{Band\ 4}))$$

The NDVI output was then subtracted or masked against the NDBI binary layer. By isolating and removing pixels with a strong soil/vegetation spectral signature, the final urban footprint was significantly refined, ensuring that the subsequent area extraction accurately represented true concrete infrastructure.

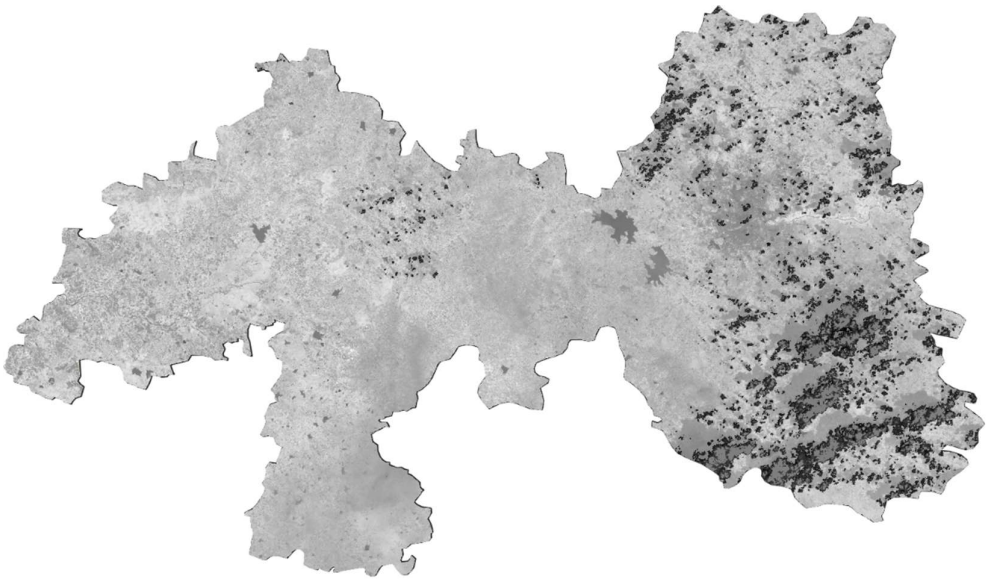

Figure 5: NDVI Map for the year 2000

Figure 6: Barren Soil data adjusted for the year 2000.

5.4 Area Extraction and Raster-to-Vector Conversion

Once the binary built-up masks were generated for all five observations years, the next step was to extract precise, quantifiable area statistics.

Because Landsat imagery has a spatial resolution of 30 meters, each individual pixel represents an area of 900 square meters on the ground (30m × 30m = 900 m²). To calculate the total built-up area for the Greater Hyderabad region, the Raster Layer Unique Values Report tool was utilized in QGIS.

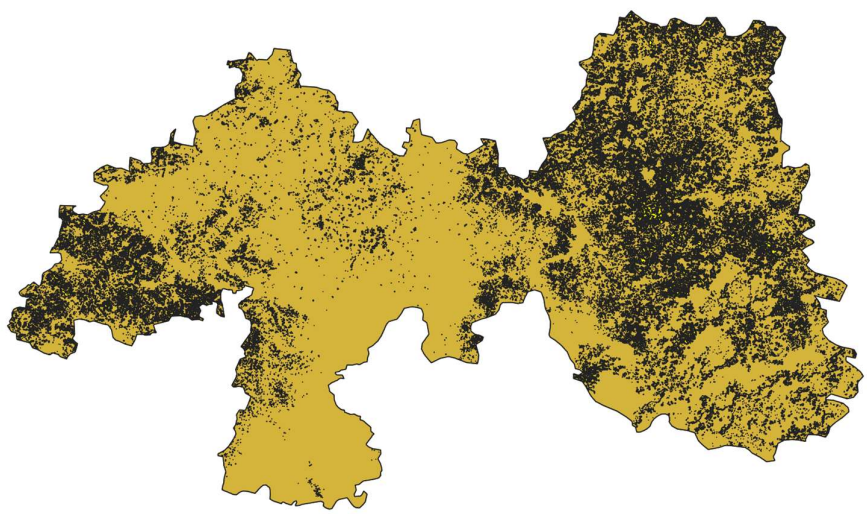

Figure 7: Vectorised output of Built-up area for the year 2000

This tool analysed the binary NDBI output and generated an HTML report containing the exact pixel count for the built-up class (value = 1). The pixel counts were then converted into square kilometres (sq. km) using the following formula:

$$Area\ (sq.km) = (Pixel\ Count\ \times 900)\ /\ 1000000$$

Following the quantitative extraction, the raster masks were subjected to a Raster-to-Vector conversion (Polygonize tool in QGIS). This allowed the pixelated data to be transformed into smooth vector shapefiles, facilitating the creation of the final high-resolution cartographic layouts.

5.5 Quantitative Time-Series Forecasting (Python ML)

While historical spatial data is crucial for analysing past development trends, predicting future urban sprawl requires quantitative forecasting. To achieve this, the area calculations derived from QGIS were exported into a Python environment (Jupyter Notebook) for predictive modelling.

A supervised Machine Learning approach, specifically a Simple Linear Regression Model, was implemented using the Scikit-Learn library. The model was trained using the historical observation years (2000, 2010, 2015, 2020, 2024) as the independent variable (***X***) and the corresponding built-up area (sq. km) as the dependent variable (***Y***).

The objective of the model was to identify the underlying linear trajectory of urban expansion and project this mathematical trend into the future to forecast the exact built-up area for the year 2030.

2. Dataset Preparation

The dataset consists of:

- Years: 2000, 2010, 2015, 2020, 2024
- Built-up area (sq km) corresponding to each year

The data is structured for regression analysis.

```
years = np.array([2000, 2010, 2015, 2020, 2024]).reshape(-1, 1)
area_sqkm = np.array([213.2667, 283.1418, 452.7, 581.9, 695.3])

years, area_sqkm
```

```
(array([[2000],
        [2010],
        [2015],
        [2020],
        [2024]]),
 array([213.2667, 283.1418, 452.7   , 581.9   , 695.3   ]))
```

3. Model Development – Linear Regression

We use a Linear Regression model to understand the relationship between:

- Independent Variable (X): Year
- Dependent Variable (Y): Built-up Area

The model fits a straight line: [Y = mX + c]

```
model = LinearRegression()
model.fit(years, area_sqkm)

print("Slope (Growth Rate):", model.coef_[0])
```

Figure 8: Python Jupyter Notebook snapshot demonstrating the Scikit-Learn Linear Regression model training and 2030 built-up area prediction.

5.6 Challenges and Troubleshooting

Throughout the execution of this analysis, several technical challenges required active troubleshooting and methodology adjustments:

1. Data Gaps and Sensor Failures: The project initially aimed for a strict 5-year interval (i.e., 2000, 2005, 2010). However, attempting to acquire 2005 imagery revealed that the Landsat 7 satellite suffered an SLC-off hardware failure in May 2003, resulting in severe data striping. With no suitable Landsat 5 alternative available for the region, the 2005 dataset was abandoned to protect the mathematical accuracy of the final calculations.
2. Barren Soil Spectral Confusion (2000 – 2015): The initial attempt to map urban growth using the standard NDBI formula resulted in a significant overestimation of built-up areas for the earlier years. Because dry, barren agricultural soil shares a highly similar Shortwave-Infrared (SWIR) reflectance signature with concrete infrastructure, large tracts of non-urban land were falsely classified as built-up pixels.
3. Limitations of NDVI Masking and Cloud Inconsistencies: To troubleshoot the barren soil error, an algorithmic mask utilizing the Normalized Difference Vegetation Index (NDVI) was implemented to mathematically filter out false positives. While early iterations of this mask struggled with persistent cloud-cover inconsistencies and atmospheric scattering, iterative adjustments to the Raster Calculator thresholding successfully isolated the true concrete infrastructure. This refined NDVI-NDBI masking approach was subsequently applied across all five observation years (2000–2024) to generate a clean, reliable dataset for precise ML forecasting.

6. RESULTS

The execution of the automated QGIS time-series methodology yielded significant quantitative data regarding the spatial expansion of Greater Hyderabad. The results are divided into historical area statistics (2000–2024) and the predictive Machine Learning forecast for 2030.

6.1 Built-Up Area Extraction Maps

The primary output of the QGIS spatial analysis is a series of binary extraction maps that visually document the sprawl of concrete infrastructure across the region. In the early observation years (2000 and 2010), the built-up area was heavily concentrated within the historic urban core of the Hyderabad District, with minimal structural presence in the surrounding peripheries.

However, the sequential extraction maps for 2015, 2020, and 2024 illustrate a dramatic shift in developmental patterns. A massive outward radiation of urban infrastructure is visible, particularly toward the western and northwestern zones of the Ranga Reddy district. This visual densification perfectly aligns with the establishment and rapid expansion of major IT and financial corridors, such as HITEC City and Gachibowli, confirming the spatial shift from a centralized city to an expansive metropolitan agglomeration.

Figure 9: Spatiotemporal progression of built-up urban area extraction maps across the Greater Hyderabad Metropolitan Region (2000–2024).

6.2 Quantitative Area Statistics (2000 – 2024)

While visual maps provide spatial context, statistical quantification is required for rigorous trend analysis. Following the QGIS pixel extraction and the successful application of the NDVI-based algorithmic masking across all observation years to bypass environmental noise, the absolute built-up area for Greater Hyderabad was tabulated.

The data reveals an aggressive, sustained trajectory of urban sprawl over the 24-year observation period. Between 2000 and 2024, the region experienced a continuous compounding increase in concrete infrastructure, converting vast tracts of barren and agricultural land into permanent urban settlements.

Observation Year	Total Built-Up Area (sq. km)
2000	213.2
2010	283.1
2015	452.7
2020	581.9
2024	695.3

Table 2: Time-Series Quantification of built-up urban area in Greater Hyderabad (2000 – 2024)

6.3 Machine Learning Urban Expansion Forecast (2030)

The historical area statistics were utilized as the training dataset for the Scikit-Learn Simple Linear Regression model. The model analysed the temporal relationship between the independent variable (Year) and the dependent variable (Built-up Area).

The mathematical output of the Linear Regression model demonstrated a highly positive correlation, confirming that Greater Hyderabad is experiencing exponential, rather than linear, infrastructure growth.

Upon passing the target year **2030** into the trained model's **predict()** function, the algorithm forecasted that the built-up area of Greater Hyderabad will expand to approximately **780.78** sq. km by the year 2030.

This represents a projected increase of over 85.48 sq. km of new concrete infrastructure between 2024 and 2030. Spatially, this growth is highly likely to continue radiating westward from the historic core into the Ranga Reddy district, driven by the saturation of the central IT corridors.

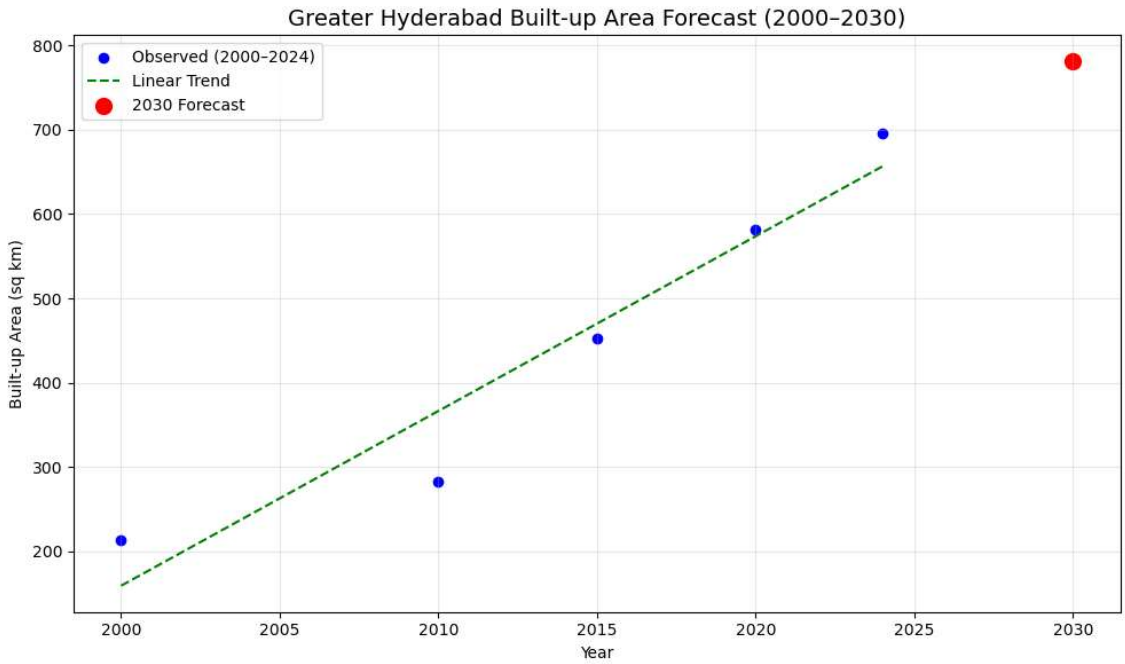

Figure 10: Linear Regression predictive model forecasting the built-up area of Greater Hyderabad for the year 2030.

7. CONCLUSION

This project successfully quantified and modelled the massive spatiotemporal urban expansion of the Greater Hyderabad Metropolitan Region over a 24-year period (2000 - 2024). By leveraging remote sensing spectral indices - specifically the Normalized Difference Built-up Index (NDBI) coupled with NDVI-based algorithmic masking - the study accurately isolated concrete infrastructure from highly reflective barren soil and agricultural land.

Furthermore, transitioning from purely historical spatial mapping to predictive Machine Learning forecasting demonstrated the advanced analytical potential of GIS when integrated with Python. The Scikit-Learn Simple Linear Regression model effectively identified the region's mathematical trajectory of growth, forecasting continued, aggressive horizontal sprawl approaching 2030. Spatially, this expansion is projected to push deeper into the Ranga Reddy district as the central IT corridors (HITEC City and Gachibowli) reach maximum saturation.

Ultimately, this rigorous methodology, combining custom QGIS raster thresholding with NDVI-based algorithmic masking to bypass atmospheric and sensor anomalies - resulted in a highly robust predictive model. The historical data demonstrates a strong, continuous increasing trend in built-up area, and the Linear Regression model estimates a consistent, aggressive growth rate projecting outward to 2030. However, it is important to acknowledge that this relies on a simple linear forecasting methodology. Real-world urban expansion is highly complex and heavily influenced by external variables such as municipal policy changes, macroeconomic factors, planned infrastructure development, and environmental constraints. For improved forecasting in future studies, advanced predictive architectures, such as Polynomial Regression, ARIMA time-series modelling, or deep learning Machine Learning algorithms (e.g., Random Forests or Neural Networks) could be explored. Despite these limitations, the current quantitative insights serve as a highly valuable, data-driven baseline for urban planners, policymakers, and real-estate developers tasked with managing sustainable infrastructure scaling in one of India’s fastest-growing metropolitan hubs.

8. REFERENCES

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